

SonoSense

Detailed Technical Architecture

AI-Powered Liver Fibrosis Staging from Routine Abdominal Ultrasound

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PROTOTYPE NOTICE: SonoSense is a research prototype. It is not a certified medical device and has not received regulatory clearance from CDSCO, FDA, or CE. Outputs must not be used for clinical diagnosis or patient care decisions.

01 SYSTEM OVERVIEW

1.1 Purpose and Scope

SonoSense is a multimodal deep learning system designed to automatically stage liver fibrosis from F0 to F4 using standard B-mode abdominal ultrasound images combined with routine clinical laboratory values. It targets the primary care and radiology settings where MASLD patients most commonly present, deploying within existing PACS workstation pipelines without requiring new hardware, specialist operators, or changes to patient workflows.

This document describes the complete technical architecture of SonoSense: the data pipeline, preprocessing stack, model architecture (segmentation branch + classification branch + fusion layer), explainability module, inference pipeline, deployment targets, and evaluation framework.

1.2 High-Level System Architecture

SonoSense is structured as a five-layer system:

- Data Ingestion & Preprocessing — DICOM handling, image normalisation, lab value parsing and imputation
- Segmentation Branch — Attention U-Net for liver boundary delineation
- Classification Branch — EfficientNet-B4 / ResNet50 ensemble CNN for fibrosis feature extraction
- Multimodal Fusion Layer — Cross-attention fusion of image embeddings with clinical tabular features
- Output & Explainability Layer — Calibrated classification head, Grad-CAM heatmap, SHAP feature attribution, structured report generation

Layer	Component	Primary Technology	Output
1 — Ingestion	DICOM parser + preprocessor	pydicom, OpenCV	Normalised PNG tensors
2 — Segmentation	Liver boundary U-Net	Attention U-Net (EfficientNet encoder)	Binary liver mask + ROI crop
3 — Classification	Fibrosis CNN encoder	EfficientNet-B4 + ResNet50 ensemble	2048-dim image embedding
4 — Fusion	Multimodal attention fusion	Cross-attention + MLP tabular encoder	Fused 512-dim representation
5 — Output	Calibrated head + explainability	Softmax + temperature scaling + Grad-CAM + SHAP	F0–F4 grade, heatmap, report

02 DATA PIPELINE & PREPROCESSING

2.1 Input Data Specification

SonoSense accepts two data streams at inference time: the ultrasound image and a structured clinical metadata object containing laboratory values and patient demographics. Both streams are required; however, missing lab values are handled via learned imputation (see Section 2.3).

2.1.1 Imaging Input

- Format: DICOM (.dcm) — converted internally to PNG for model processing
- Modality: B-mode (2D grayscale) abdominal ultrasound

- Required views: Right lobe subcostal view (primary); intercostal view (secondary, optional)
- Frame selection: If a multi-frame DICOM cine clip is provided, a quality scoring model (sharpness + liver coverage heuristic) automatically selects the best 3 frames for ensemble averaging
- Resolution: Accepted from 256×256 up to 1024×1024; internally resized to 512×512 before model ingestion
- Machine vendor: Vendor-agnostic — trained on images from Siemens, GE, Philips, and Mindray machines to ensure cross-vendor generalisation

2.1.2 Clinical Input (Tabular Stream)

Variable	Type	Required?	Clinical Rationale
AST (U/L)	Continuous	Yes	Hepatocyte injury; AST/ALT ratio rises with fibrosis stage
ALT (U/L)	Continuous	Yes	Hepatocyte injury marker; part of FIB-4 formula
Platelet count (x10 ⁹ /L)	Continuous	Yes	Falls with portal hypertension; strongest single lab predictor
Age (years)	Continuous	Yes	Part of FIB-4; fibrosis risk increases with age
GGT (U/L)	Continuous	Recommended	Sensitive to both steatosis and fibrosis progression
Albumin (g/dL)	Continuous	Recommended	Drops in advanced fibrosis as synthetic function declines
HbA1c (%)	Continuous	Optional	Reflects metabolic dysfunction driving fibrosis
BMI (kg/m ²)	Continuous	Optional	Strongest clinical predictor in gradient-boosted models
ALP (U/L)	Continuous	Optional	Cholestasis marker; useful for advanced disease
Bilirubin (mg/dL)	Continuous	Optional	Secondary signal for F3–F4 and cirrhosis
Type 2 diabetes (Y/N)	Binary	Optional	Independently doubles fibrosis risk
Sex (M/F)	Binary	Optional	Sex-specific progression patterns

2.2 DICOM Ingestion Pipeline

Raw DICOM files are parsed using pydicom. The pipeline performs the following steps in sequence:

- Metadata extraction: Patient ID, study date, ultrasound machine vendor/model, and acquisition parameters are extracted and logged for audit trail
- Pixel array extraction: The raw pixel array is extracted and converted to uint8 grayscale
- Photometric interpretation check: MONOCHROME1 images are inverted; RGB images are converted to grayscale via luminosity weighting
- Probe region detection: A heuristic crops out the on-screen annotation overlay using connected-component analysis on high-entropy border regions
- Fan-mask removal: Ultrasound fan-shaped scanning region is isolated from background using Otsu thresholding; only the scan region is passed downstream

2.3 Image Normalisation & Augmentation

2.3.1 Normalisation

- CLAHE (Contrast Limited Adaptive Histogram Equalisation): Applied with clip limit 2.0, tile grid 8×8 — enhances echotexture visibility without amplifying noise
- Z-score normalisation per image: Mean subtracted, divided by standard deviation — accounts for inter-machine gain differences

- Resize to 512×512: Bicubic interpolation

2.3.2 Training-Time Augmentation

To address the limited size of ultrasound fibrosis datasets and improve generalisation across machine vendors:

- Geometric: Random horizontal flip ($p=0.5$), random rotation $\pm 15^\circ$, random elastic deformation ($\sigma=10$, $\alpha=100$)
- Intensity: Random brightness/contrast jitter ($\pm 20\%$), random Gaussian noise (σ in $[0, 0.05]$), random gamma correction (γ in $[0.8, 1.2]$)
- Ultrasound-specific: Random speckle noise injection, random shadowing artifact simulation (vertical dark bands), random gain simulation
- Mixup: Applied with $\alpha=0.2$ between same-stage examples only — prevents boundary blur between adjacent stages

2.4 Missing Lab Value Imputation

At inference, it is common for one or more optional lab values to be absent. SonoSense handles this in two ways:

- Median imputation (baseline): Missing values are replaced by the population median from the training cohort, stratified by age group and sex
- Learned missingness mask (advanced): A binary missingness mask vector is concatenated to the tabular input tensor, allowing the MLP tabular encoder to learn imputation-aware representations — the model learns to down-weight imputed values implicitly

The minimum viable input set is: one ultrasound image + AST + ALT + platelet count + age. All other features improve performance but are never required.

03 SEGMENTATION BRANCH — LIVER BOUNDARY DELINEATION

3.1 Architecture: Attention U-Net with EfficientNet-B4 Encoder

The segmentation branch produces a binary liver mask from the preprocessed ultrasound image. This mask serves two purposes: (1) it gates the classification branch to analyse only liver parenchyma, excluding surrounding tissue and artefacts; (2) it provides the spatial canvas onto which the Grad-CAM heatmap is projected for clinical display.

3.1.1 Encoder

- Backbone: EfficientNet-B4 pretrained on ImageNet, adapted for single-channel greyscale input by averaging the three-channel first-layer weights
- Feature pyramid: Encoder produces feature maps at 5 spatial scales — 256×256, 128×128, 64×64, 32×32, 16×16 — passed to decoder via skip connections
- Rationale: EfficientNet-B4 achieves highest sensitivity (0.85) among CNN backbones tested on ultrasound liver classification tasks, with fewer parameters than ResNet50 or DenseNet

3.1.2 Decoder with Attention Gates

- Architecture: Standard U-Net decoder with transposed convolution upsampling blocks at each scale
- Attention gates: Inserted at every skip connection — each gate computes a soft attention weight map from the gating signal (decoder) and encoder feature map, suppressing background and noise while emphasising liver boundary regions

- Multi-scale dilated convolutional attention module (MDCAM): Placed at the bottleneck to capture global contextual information across varying liver shapes and sizes
- Boundary decoder: A separate lightweight decoder branch explicitly supervised on liver boundary masks during training — sharpens capsule edge delineation

3.1.3 Training the Segmentation Branch

- Loss function: Combined Dice loss + Binary Cross-Entropy loss ($\lambda=0.5$ each) — Dice handles class imbalance; BCE ensures pixel-level precision
- Label source: Liver masks from Saudi NAFLD OSF dataset (manually annotated) + auto-generated pseudo-labels from a pre-trained nnU-Net applied to additional unlabelled scans
- Target metric: Dice Similarity Coefficient (DSC) ≥ 0.94 on held-out test set — matching Attention U-Net published benchmarks of 0.9468

3.2 ROI Cropping

After segmentation, the bounding box of the predicted liver mask is extracted with a 5% padding margin. The original image is cropped to this bounding box and resized to 384×384 for input to the classification branch. This ensures the CNN operates entirely within liver parenchyma, improving sensitivity to echotexture changes and eliminating irrelevant background signal.

04 CLASSIFICATION BRANCH — FIBROSIS FEATURE EXTRACTION

4.1 Architecture: EfficientNet-B4 + ResNet50 Ensemble

The classification branch takes the liver-cropped ROI image as input and produces a dense image embedding capturing fibrosis-relevant features: echotexture coarseness, increased echogenicity, posterior echo attenuation, surface nodularity, and capsule irregularity.

4.1.1 Dual Backbone Ensemble

A 2025 Scientific Reports paper demonstrated that fusing EfficientNetB0 and ResNet50 features achieved 99.45% accuracy on liver fibrosis staging — significantly outperforming either backbone alone. SonoSense adopts this ensemble principle with EfficientNet-B4 (stronger than B0) and ResNet50:

Backbone	Parameters	Strengths	Output embedding dim
EfficientNet-B4	19M	Best sensitivity on ultrasound; efficient compound scaling; fewer FLOPs	1792-dim
ResNet50	25M	Strong residual feature learning; well-studied on medical images; complementary texture features	2048-dim

- Feature concatenation: After global average pooling, the 1792-dim and 2048-dim embeddings are concatenated to form a 3840-dim combined image embedding
- Dimensionality reduction: A linear projection layer with dropout ($p=0.3$) reduces this to a 1024-dim image feature vector for fusion
- Weight sharing with segmentation encoder: The EfficientNet-B4 encoder weights are partially shared with the segmentation branch — layers 1–5 are shared (multi-task learning), layers 6–9 are task-specific — reducing total parameter count by ~35%

4.1.2 Hybrid CNN-Transformer Attention

The classification branch optionally incorporates a lightweight Transformer self-attention module applied to the patch-level feature grid from EfficientNet-B4 (before global average pooling). This allows the model to capture long-range spatial dependencies — correlating surface irregularity at the liver capsule with internal echotexture changes — which purely convolutional models miss.

- Implementation: 4-head self-attention on a 16×16 spatial grid of patch embeddings from the EfficientNet-B4 penultimate feature map
- Parameter cost: +1.8M parameters — small relative to the backbone
- Ablation finding: Hybrid CNN-Transformer architectures consistently outperform pure CNN models by 1.5–3% AUROC on hepatic ultrasound classification tasks

4.2 Pre-training Strategy

Both backbones are initialised with ImageNet weights and then fine-tuned in two stages:

- Stage 1 — Ultrasound domain adaptation: Pre-trained on a large corpus of unlabelled abdominal ultrasound images using self-supervised contrastive learning (SupCon loss). A 2025 PMC paper (LivSCP) demonstrated that supervised contrastive pretraining on liver ultrasound achieved AUROC values of 0.989–1.000 across all fibrosis stages.
- Stage 2 — Task-specific fine-tuning: Fine-tuned on labelled fibrosis staging data (Saudi OSF + Kaggle + Korean hospital datasets) with a cosine annealing learning rate schedule (initial LR: 1e-4, min LR: 1e-6, warmup: 5 epochs)

05 TABULAR ENCODER — CLINICAL FEATURE PROCESSING

5.1 MLP Tabular Encoder

The clinical laboratory values and patient metadata are encoded by a dedicated multi-layer perceptron (MLP) that produces a 256-dim clinical embedding, which is then fused with the 1024-dim image embedding in the fusion layer.

5.1.1 Input Feature Engineering

Before passing lab values to the MLP, four derived features are computed and appended to the raw lab vector:

- FIB-4 score: $(\text{Age} \times \text{AST}) / (\text{Platelet} \times \sqrt{\text{ALT}})$ — the current clinical standard, fed in so the model can learn when it is wrong
- APRI score: $(\text{AST} / 40) \times 100 / \text{Platelet}$ — independent fibrosis signal
- AST/ALT ratio: Rises above 1.0 in advanced fibrosis, providing a simple but powerful discriminator
- Albumin/Globulin ratio: Derived from albumin and total protein — imbalance reflects synthetic liver dysfunction

This gives a total of up to 16 input features (12 raw + 4 derived), with a missingness mask of equal length concatenated, yielding a 32-dim input vector to the MLP.

5.1.2 MLP Architecture

- Input layer: 32 neurons (16 features + 16 missingness flags)
- Hidden layers: 128 → 256 → 256 neurons with ReLU activation, batch normalisation, and dropout (p=0.25) at each layer
- Output: 256-dim clinical embedding
- Regularisation: L2 weight decay (lambda=1e-4) to prevent overfitting on small tabular datasets

06 MULTIMODAL FUSION LAYER

6.1 Cross-Attention Fusion

The 1024-dim image embedding and the 256-dim clinical embedding are combined using a cross-attention mechanism — the image embedding attends over clinical features and vice versa, producing a jointly modulated 512-dim fused representation.

6.1.1 Fusion Architecture

- Query: Clinical embedding (256-dim), projected to 512-dim
- Key / Value: Image embedding (1024-dim), projected to 512-dim
- Attention: Single-head cross-attention; attention weights are logged per inference for interpretability — they reveal how much each clinical variable modulated the image prediction
- Concatenation fallback: If attention training is unstable during early runs, a simpler concatenation + linear projection layer ($1024 + 256 \rightarrow 512$) is used as the fusion mechanism — both are implemented and switchable via config flag

6.2 Why Fusion Outperforms Single-Stream Models

Published evidence confirms the multimodal advantage: a Nature Communications 2026 paper integrating patient history, routine labs, and ultrasound features achieved AUROC 0.90 for advanced fibrosis — outperforming FIB-4 alone and ultrasound-only models. The cross-attention mechanism specifically allows the model to condition its image interpretation on clinical context — for example, treating an ambiguous echotexture reading differently when platelet count is normal vs. critically low.

07 OUTPUT LAYER — CLASSIFICATION, CALIBRATION & EXPLAINABILITY

7.1 Classification Head

- Architecture: Two-layer MLP on the 512-dim fused representation $\rightarrow$ 5-class softmax output (F0, F1, F2, F3, F4)
- Ordinal constraint: A cumulative link model (CLM) layer is applied on top of raw logits to enforce ordinality — the model cannot predict a higher probability for F2 than F0 while simultaneously predicting a low probability for F1
- Multi-threshold outputs: Five separate binary classifiers are derived from the main model to support specific clinical decision thresholds described in the evaluation framework

7.2 Confidence Calibration

Raw neural network softmax probabilities are systematically overconfident and must be calibrated before clinical use. SonoSense applies:

- Temperature scaling: A single scalar parameter T is learned on the validation set — logits are divided by T before softmax, expanding the probability distribution. Target: Expected Calibration Error (ECE) < 0.05
- Platt scaling (fallback): A logistic regression layer trained on validation set predictions provides an alternative calibration approach for binary threshold outputs
- Reliability diagrams: Generated on the test set and included in the evaluation report — bins of 0.1 width across the $[0,1]$ confidence range, plotting mean predicted probability vs. observed frequency

7.3 Uncertainty Quantification

A single deterministic prediction is insufficient for clinical trust. SonoSense provides two uncertainty measures:

- MC Dropout: At inference, dropout is kept active and 20 forward passes are run. The mean of predictions is the final output; the standard deviation is the epistemic uncertainty estimate. High uncertainty outputs ($\sigma > 0.15$) are flagged with a 'Low Confidence — Recommend Clinical Review' warning
- Prediction interval: The 10th–90th percentile range across MC Dropout passes is reported alongside the point estimate — e.g., 'F2 (84% confidence, range: F1–F3)'

7.4 Explainability — Grad-CAM Heatmap

Gradient-weighted Class Activation Mapping (Grad-CAM) is applied to the EfficientNet-B4 backbone to produce a spatial heatmap indicating which liver regions drove the predicted fibrosis stage. Published liver fibrosis staging studies applying Grad-CAM achieved AUROCs of 0.92, 0.89, and 0.88 for significant fibrosis, advanced fibrosis, and cirrhosis respectively.

7.4.1 Heatmap Generation Pipeline

- Target layer: The final convolutional layer of EfficientNet-B4 (before global average pooling)
- Gradient computation: Gradients of the predicted class score with respect to the target layer activations are computed and global-average-pooled to produce channel weights
- Activation weighting: Channel weights multiply the activation maps and are ReLU-clipped to retain only positive contributions
- Upsampling: The resulting low-resolution activation map is bilinearly upsampled to match the original image resolution (512×512)
- Overlay: The heatmap is blended onto the original ultrasound image ($\alpha=0.45$) using a jet colourmap — red/yellow regions indicate high fibrosis signal; blue/cool regions indicate low contribution
- Masking: The heatmap is zero-masked outside the U-Net liver segmentation boundary — only activations within the liver region are shown

7.4.2 Heatmap Validation (Faithfulness Testing)

- Occlusion test: High-activation regions are progressively occluded; model confidence must drop proportionally — if not, the heatmap is flagged as non-faithful and Grad-CAM++ is used instead
- Pointing game: The peak activation must fall within a hepatologist-annotated region of interest — target accuracy $\geq 70\%$ on held-out annotated subset
- IoU with expert annotations: Target Intersection over Union ≥ 0.50 on a 100-image subset annotated by two hepatologists

7.5 SHAP Feature Attribution (Tabular Explanation)

To explain the contribution of individual clinical variables to each prediction, SHAP (SHapley Additive Explanations) values are computed for the tabular MLP encoder output. This tells the clinician: 'platelet count of $98 \times 10^9/L$ contributed +0.23 toward an F3 prediction, while normal ALT contributed -0.11.' SHAP values are displayed in the structured report as a horizontal bar chart ordered by contribution magnitude.

7.6 At-Risk MASH Flag and Referral Logic

Any prediction of F2 or above triggers the At-Risk MASH flag automatically. The referral recommendation is generated as follows:

Predicted Stage	Confidence	Action Generated
F0–F1	$\geq 70\%$	No referral. Reassess in 12 months if metabolic risk factors present.
F0–F1	$< 70\%$	Indeterminate result. Recommend FIB-4 repeat or hepatology review.
F2	$\geq 70\%$	At-Risk MASH flag. Hepatology referral recommended. Consider Resmetirom eligibility assessment.
F3	Any	Advanced fibrosis. Urgent hepatology referral. Varices screening recommended.
F4	Any	Cirrhosis. Immediate hepatology referral. HCC surveillance protocol.

08 STRUCTURED REPORT GENERATION

8.1 Report Contents

Each SonoSense inference generates a structured clinical report containing:

- Patient demographics and scan metadata (date, machine, operator ID)
- Three-panel image: original ultrasound | liver segmentation overlay | Grad-CAM heatmap
- Fibrosis grade: F0–F4 with confidence percentage and prediction interval
- Uncertainty flag: 'Low Confidence' warning if MC Dropout $\sigma > 0.15$
- SHAP bar chart: Top 5 contributing clinical variables
- FIB-4 corroboration: Computed FIB-4 score displayed alongside SonoSense grade — flagged as 'consistent' or 'discordant' (discordance triggers a review prompt)
- At-Risk MASH flag and recommended action
- Disclaimer: Prototype research tool notice — not for clinical use

8.2 Export Formats

- PDF: Self-contained clinical summary for printing or EMR attachment
- HL7 FHIR R4 DiagnosticReport resource: Machine-readable, structured for direct EMR ingestion (Epic, Cerner, OpenMRS)
- JSON: Raw inference output for API consumers and downstream analytics

09 DEPLOYMENT ARCHITECTURE

9.1 Deployment Targets

9.1.1 PACS Plugin (Primary Deployment)

SonoSense deploys as a DICOM node within the hospital PACS network. When a radiologist opens an abdominal ultrasound study for reporting, SonoSense automatically receives the DICOM series via DICOM C-STORE, runs inference, and pushes the structured report back as a DICOM SR (Structured Report) and secondary capture image. The radiologist sees the heatmap overlay and grade within their existing reporting interface with zero additional steps.

9.1.2 REST API (Cloud / EMR Integration)

- Endpoint: POST /v1/infer — accepts multipart form data with DICOM file + JSON clinical data

- Response: JSON inference result + base64-encoded heatmap PNG + FHIR DiagnosticReport
- Authentication: OAuth 2.0 bearer tokens; all payloads encrypted in transit (TLS 1.3)
- SLA: Inference latency < 8 seconds end-to-end (preprocessing + segmentation + classification + report generation) on T4 GPU

9.1.3 SaaS Dashboard (Clinic Portal)

Gastroenterology and primary care clinics without PACS infrastructure access SonoSense via a web dashboard. Clinicians upload the ultrasound image and enter lab values manually or via CSV import. Results are returned within 10 seconds and stored in an audit-logged patient record within the dashboard.

9.2 Infrastructure

Component	Technology	Notes
Model serving	FastAPI + Uvicorn on Docker	Containerised inference; horizontal scaling via Kubernetes
GPU inference	NVIDIA T4 / A10G	ONNX Runtime for optimised inference; TorchScript export
Model registry	MLflow	Version tracking, experiment logging, A/B testing between model versions
DICOM handling	Orthanc PACS + pydicom	Receives and sends DICOM via C-STORE and C-FIND
Database	PostgreSQL	Patient records, audit logs, prediction history
Storage	AWS S3 / on-prem MinIO	DICOM archives, model artefacts, report PDFs
Monitoring	Prometheus + Grafana	Inference latency, confidence distribution drift, error rates
Security	TLS 1.3, AES-256 at rest	HIPAA-aligned data handling; de-identification pipeline for research exports

10 TRAINING DATASETS & EXPERIMENTAL SETUP

10.1 Dataset Stack

Dataset	Images/Patients	Labels	Lab Data	Access
Saudi NAFLD (OSF: osf.io/c2yg8)	10,352 images / 384 pts	NAS F0–F4 + steatosis (biopsy-proven)	No	Open access
BEHSOF (Figshare)	~500 images / 113 pts	Steatosis + fibrosis grade	Yes (full labs)	Open access
Kaggle Liver Fibrosis (vibhingupta028)	~2,500 images	F0–F4 stage	No	Open access
Korean Hospital (Park et al. 2024)	7,920 images / 933 pts	METAVIR F0–F4 (biopsy/hepatectomy)	No	Request from authors
NIDDK NASH CRN	2,500+ pts	NAS biopsy gold standard + full labs	Yes (complete)	Apply via NIDDK repository

10.2 Data Split Strategy

- Training set: 65% — model learns weights
- Validation set: 15% — hyperparameter tuning, early stopping, calibration
- Internal test set: 20% — reported performance metrics, never touched during training
- External validation: Separate hospital dataset not used in any training phase — essential for regulatory-grade claims

Patient-level splits are enforced: all images from the same patient appear in the same split only, preventing data leakage from multi-frame studies.

10.3 Training Configuration

Hyperparameter	Value	Rationale
Optimiser	AdamW (beta1=0.9, beta2=0.999)	Weight decay decoupling prevents feature co-adaptation
Learning rate	1e-4 (backbone), 1e-3 (head)	Differential LR: slower fine-tuning for pretrained layers
LR schedule	Cosine annealing with warm restart	Avoids sharp loss valleys; improves generalisation
Batch size	32 (image) / 128 (tabular)	Larger tabular batches for stable BN statistics
Epochs	100 max with early stopping (patience=15)	Prevents overfitting on small ultrasound datasets
Loss — segmentation	Dice + BCE (lambda=0.5 each)	Handles class imbalance; pixel precision
Loss — classification	Weighted cross-entropy + ordinal penalty	Higher weight on F3/F4 minority classes; penalises large stage errors
Class weights	Inverse frequency weighting	F3 and F4 are underrepresented — upweighted by factor 3–5x
Hardware	2x NVIDIA A100 40GB	Multi-GPU training via PyTorch DDP; ~18h for full pipeline

11 EVALUATION FRAMEWORK

11.1 Primary Metrics

Metric	Target	Clinical Threshold
AUROC — F2+ (significant fibrosis)	≥ 0.88	Regulatory-grade performance for staging tool
AUROC — F3+ (advanced fibrosis)	≥ 0.90	Strong discrimination for urgent referral decisions
Sensitivity — F2+	$\geq 85\%$	Missing an at-risk patient is the primary harm to avoid
Specificity — F2+	$\geq 78\%$	Limits unnecessary referrals and patient anxiety
AUPRC — F2+	≥ 0.80	Accounts for class imbalance in real-world prevalence
Quadratic Weighted Kappa	≥ 0.65	Substantial agreement with biopsy gold standard
Expected Calibration Error (ECE)	< 0.05	Well-calibrated confidence scores for clinical trust
Heatmap IoU	≥ 0.50	Heatmap reliably highlights radiologist-identified regions

11.2 Benchmark Comparisons

All reported metrics are compared head-to-head against:

- FIB-4 score alone — current standard of care in primary care
- FibroScan (VCTE) — current best non-invasive tool
- Radiologist reading of the same ultrasound without SonoSense
- Radiologist reading with SonoSense — to demonstrate augmentation benefit

The most clinically compelling result is: SonoSense alone $\geq$ FibroScan, and Radiologist + SonoSense $>$ Radiologist alone, measured by both AUROC and Net Reclassification Improvement (NRI) vs. FIB-4.

11.3 Clinical Impact Metrics

- Net Reclassification Improvement (NRI) vs. FIB-4: Proportion of patients correctly moved from indeterminate or wrong category to correct stage
- Integrated Discrimination Improvement (IDI): Improvement in predicted probability separation between true positives and negatives vs. FIB-4
- Number Needed to Screen (NNS): How many ultrasounds SonoSense reads to correctly identify one at-risk patient missed by FIB-4 alone

12 DEVELOPMENT ROADMAP

Phase	Timeline	Milestones
Phase 1 — Baseline	Now → Aug 2026	Saudi OSF + Kaggle data prep; baseline EfficientNet-B4 image-only classifier; internal test AUROC ≥ 0.80 for F2+
Phase 2 — Multimodal	Aug → Oct 2026	Tabular encoder integration; cross-attention fusion; BEHSOF lab data incorporated; target AUROC ≥ 0.87
Phase 3 — Explainability	Sep → Oct 2026	Attention U-Net segmentation branch; Grad-CAM pipeline; SHAP attribution; heatmap IoU validation
Phase 4 — Prototype Launch	Early Nov 2026	Full pipeline integrated; SaaS dashboard live; structured report generation; disclaimer-compliant public prototype
Phase 5 — Clinical Validation	Nov 2026 → 2027	Partner hospital data collection; external validation cohort; NRI/IDI vs. FIB-4; ethics board submission
Phase 6 — Regulatory	2027+	CDSCO Class B medical device submission; CE Mark technical file; FDA De Novo pathway assessment

IMPORTANT NOTICES

PROTOTYPE NOTICE: SonoSense is a research prototype. It is not a certified medical device and has not received regulatory clearance from CDSCO, FDA, or CE. Outputs must not be used for clinical diagnosis or patient care decisions.

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